Author: Micheal Lanham

SECTION 1

Understanding Retrieval in AI Applications

- **Retrieval bridge:** Retrieval connects unbounded external stores to the bounded context window by fetching only the most relevant pieces for each LLM call.
- **Knowledge versus memory:** Knowledge is external reference material; memory captures experiential interaction history—both use the same retrieve-then-augment pattern.
- **RAG phases:** RAG splits into ingestion (load, chunk, embed, index) and query-time retrieval (embed query, search, augment prompt, generate).
- **Semantic search:** Dense embeddings compare meaning rather than surface keywords, enabling matches like “vehicles” to “cars,” with pitfalls around factual correctness, fixed top-k, and approximate nearest-neighbor indexes.
- **Why agents need retrieval:** Models cannot answer from post-training facts unless the system supplies current documents, policies, prices, or session-specific state at query time.

SECTION 2

Vector Databases and Similarity Search

- **TF-IDF baseline:** TF-IDF builds sparse vectors from term importance and supports cosine similarity, but cannot align different words with the same concept.
- **Neural embeddings:** Pretrained embedding models map text into dense vectors where related meanings sit nearby, trained by predicting contextual structure rather than counting tokens.
- **Cosine similarity:** Cosine similarity compares vector direction and is standard for both sparse and dense retrieval, ignoring magnitude effects from document length.
- **In-memory and Chroma:** Vectors can live in arrays for experiments; Chroma DB provides a practical local collection API with distance-ranked query results.
- **Hybrid signal:** Production systems often blend BM25/keyword precision with semantic recall because neither alone covers codes, jargon, duplicates, and freshness.

SECTION 3

Building Practical RAG Knowledge Agents

- **Vector-only limits:** A vector-search RAG agent grounded in ingested chunks answers well on semantic questions but may miss exact times, codes, or rare strings without complementary search.
- **Grounding:** Grounding instructions require answers to cite retrieved passages and admit when context is insufficient, reducing hallucination from parametric model knowledge.
- **Hybrid RAG agent:** Agents with separate vector and keyword tools choose or combine methods, enforcing quotes and reference markers in structured answer formats.
- **Relevance engineering:** Tables of failure modes—missed literals, jargon drift, duplicate crowding, stale facts—motivate keyword, SQL, graph, and hybrid supplements beyond pure similarity.
- **Reciprocal Rank Fusion:** RRF merges ranked lists from heterogeneous scorers by rank position, avoiding calibration problems between BM25 and cosine scores.

SECTION 4

Adding Memory to Agents with MCP

- **Architectural memory view:** Agent memory decomposes into context window, external storage, state scratchpads, and retrieval mechanisms rather than purely cognitive metaphors.
- **Graph memory via MCP:** The MCP memory server exposes tools to store entities, relationships, and observations in a graph optimized for traversing connections.
- **Hybrid memory:** Hybrid memory runs semantic search first for conversational context, then graph lookups for structured facts, writing new information to both stores.
- **Semantic, episodic, procedural:** Episodic events and procedural steps fit relational patterns; semantic augmentation generates likely future queries to improve later recall of those records.
- **Compression and forgetting:** Cluster-and-summarize compression collapses near-duplicate memories; forgetting or eviction controls noise, with regulated domains balancing retention rules against pruning risk.